ABHINAV SUJIL

Cybersecurity Engineer | Application Security (AppSec) & VAPT

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PROFESSIONAL SUMMARY

Cybersecurity engineer focused on application security and VAPT, with hands-on web testing for OWASP Top 10 issues using Burp Suite, Nmap, and manual HTTP payloads. Solo builder and security owner of H2O OMS, a multi-tenant queue and CRM platform operating in live production at an Akshaya centre serving 100+ citizens daily. Background spanning defense R&D embedded machine learning at DRDO STARC and enterprise threat detection engineering using Wazuh SIEM and Sysmon. Google Cybersecurity certified, CEH v12 in progress.

CORE TECHNICAL SKILLS

- **Penetration Testing & Security Tools:** Burp Suite, Nmap, Metasploit Framework, OWASP ZAP, SQLMap, Wireshark, Nikto.
- **Application & API Security:** OWASP Top 10, REST API Security, Multi-Tenant Row-Level Security (RLS), Session Management, Role-Based Access Control (RBAC).
- **Security Operations & Blue Team:** SIEM (Wazuh, Google Chronicle, Splunk fundamentals), Microsoft Sysmon, Windows Event Logs, Linux Auditd/Syslog, MITRE ATT&CK.
- **Frameworks & Standards:** OWASP Web Security Testing Guide (WSTG), NIST SP 800-115 (Security Testing), NIST Cybersecurity Framework (CSF), ISO/IEC 27001 Controls.
- **Programming & Engineering:** Python (Security Automation, Log Parsing), JavaScript/TypeScript (Next.js, Node.js), SQL (PostgreSQL, Prisma), Bash Scripting, C/C++ (ESP32/Embedded), Git.
- **Platforms & Infrastructure:** Kali Linux, Ubuntu Server, Windows 11 Enterprise, Cloudflare Workers, Docker, AWS/GCP Security Fundamentals.

PROFESSIONAL EXPERIENCE

SOFTMASON TECHNOLOGIES — **Bengaluru, Karnataka**
Cybersecurity Engineer — *September 2026 – Present*

- Undergoing structured engineering onboarding and application security training focusing on secure web architectures, OWASP Top 10 vulnerabilities, and API authentication flows.
- Shadowing security assessments across internal staging environments and web application modules, evaluating input validation, session management, and authorization boundaries.
- Conducting controlled vulnerability scans and reconnaissance using Nmap and Burp Suite against designated test targets to identify common misconfigurations and exposed endpoints.
- Developing Python scripts for automated log parsing and endpoint data extraction to support internal security monitoring workflows.

CYBORGMINDSET.AI — **Bengaluru, Karnataka**
AI Engineer — *October 2025 – November 2025*

- Engineered automated multi-platform workflow integrations using REST APIs and webhook connectors, eliminating manual multi-portal operational overhead.
- Audited and secured API authentication flows across distributed SaaS endpoints, enforcing proper token validation, secret key protection, and least-privilege access controls.
- Conducted security and integration testing across automated communication pipelines, identifying and mitigating unauthorized data exposure risks in outbound payloads.

DRDO – STARC (SITAR) — **Bengaluru, Karnataka**
Technical Intern – Design & Development — *April 2025 – August 2025*

- Engineered a real-time acoustic drone detection system using machine learning and deep learning models, increasing detection accuracy from 75% to 92% through spectrogram analysis and noise filtering.
- Optimized and deployed ML drone classification models onto low-power embedded hardware, achieving real-time inference latency for aerial defense monitoring.
- Conducted rigorous system verification in high-security defense R&D environments, adhering to DRDO data security, confidentiality, and air-gapped system isolation protocols.

AIROBOSOFT PRODUCTS AND SERVICES — **Bengaluru, Karnataka**
Software Developer Intern — *November 2023 – December 2023*

- Built real-time facial recognition verification software using Python and OpenCV, achieving ~90% accuracy on live video feeds for physical surveillance monitoring.
- Engineered automated alerting workflows to notify monitoring personnel immediately upon detecting unauthorized or unlisted access attempts.

SECURITY ASSESSMENTS & TECHNICAL PROJECTS

H2O OMS (Happy 2 Organise) – Multi-Tenant CRM & Queue Platform **Founder & Solo Developer**
Next.js, Node.js/Express, PostgreSQL (Prisma RLS), Cloudflare Workers, Edge Hardware *Live in Production since Jan 2026*

- Built and deployed an end-to-end multi-tenant CRM and queue management platform for Akshaya centres; live pilot currently serves 100+ citizens daily with zero unplanned downtime.
- Integrated distributed physical service terminals communicating through an edge proxy with per-device HMAC-signed request authentication.
- Executed a rigorous 22-item internal security audit on the production platform; uncovered and remediated a critical multi-tenant isolation flaw where physical client devices shared a single backend authorization key.
- Identified and patched an information disclosure vulnerability in the service health check endpoint that leaked device key hashes to unauthenticated external queries.
- Tested and hardened authorization boundaries, input sanitization routines, and session cookie security attributes (HttpOnly, Secure, SameSite) across Admin and Staff privilege tiers.
- Authored formal security assessment documentation with CVSS v3.1 severity scores and a prioritized technical remediation roadmap.

Enterprise SOC Detection Lab & Adversary Emulation **Defensive Security Lab**
Architecture: Wazuh SIEM Manager, Windows 11 Enterprise (Sysmon), Ubuntu Server (Auditd) *Lab Project*

- Architected an enterprise detection environment to simulate real-world cyber attacks mapped to the **MITRE ATT&CK framework**: LSASS credential dumping (T1003.001), obfuscated PowerShell execution (T1059.001), and security defense tampering (T1562.001).
- Authored custom Wazuh XML detection rules and correlation logic to alert on post-exploitation techniques with zero default vendor coverage.
- Published an enterprise incident response report aligned with NIST SP 800-61 Rev. 2 containing IoCs, timeline reconstruction, and endpoint attack surface reduction (ASR) baselines.

Automated Threat Intelligence & IoC Triage Engine **Python Security Tool**
Integration: VirusTotal API v3, AbuseIPDB API v2, RESTful Log Parsing *Automation Tool*

- Engineered a modular Python CLI tool to automate analyst reputation queries for suspect IP addresses, domains, and file hashes (MD5, SHA256).
- Implemented input validation, rate limiting, and an algorithmic risk-scoring engine, parsing threat telemetry directly into structured markdown triage summaries for rapid incident response.

Secure NFC-Based Access Control System **IoT & Embedded Security**
Hardware: ESP32 Microcontroller, RFID/NFC Sensors, Tamper-Evident Audit Logging *Hardware Security*

- Designed and built a physical access verification device with cryptographic card UID token validation and automated tamper alerts.
- Enforced tamper-evident, timestamped access logging to prevent unauthorized facility entry and defend against badge cloning and replay attacks.

EDUCATION

MVJ COLLEGE OF ENGINEERING **Bengaluru, Karnataka**
Bachelor of Engineering (B.E.) in Computer Science & Engineering *2021 – 2025*

Relevant Coursework: Network Security, Operating Systems, Database Management Systems, Data Structures & Algorithms, Computer Networks, Object-Oriented Programming.

CERTIFICATIONS & TRAINING

- **Certified Ethical Hacker (CEH v12)** – EC-Council (In Progress / Certification Track)
- **Google Cybersecurity Professional Certificate** – Coursera / Google (Completed Nov 2024)
- **Cybersecurity Job Simulation** – J.P. Morgan / Forage (Fraud Data Analysis, Phishing Detection)

PROFESSIONAL ACTIVITIES & SECURITY INTERESTS

- **Offensive Research:** Web Application Security, Bug Bounty Hunting, OWASP Top 10 Testing & Remediation.
- **Hands-On Labs & CTFs:** Active participant on TryHackMe, PortSwigger Web Security Academy, and offensive capture-the-flag competitions.
- **Languages:** English (Professional), Malayalam (Native), Hindi (Working proficiency).